

```

SELECT *
FROM (
    SELECT ProductID,
           ProductName,
           Category,
           Price,
           ROW_NUMBER() OVER (
               PARTITION BY Category
               ORDER BY Price DESC
           ) AS RowNum
    FROM Products
) t
WHERE RowNum <= 3;

```

Output

ProductID	ProductName	Category	Price	RowNum
1	Laptop A	Laptop	80000	1
2	Laptop B	Laptop	75000	2
3	Laptop C	Laptop	75000	3
5	Phone A	Phone	50000	1
6	Phone B	Phone	45000	2
7	Phone C	Phone	45000	3

```

SELECT *
FROM (
    SELECT ProductID,
           ProductName,
           Category,
           Price,
           ROW_NUMBER() OVER (
               PARTITION BY Category
               ORDER BY Price DESC
           ) AS RowNum
    FROM Products
) t
WHERE RowNum <= 3;

```

Output

ProductID	ProductName	Category	Price	RowNum
1	Laptop A	Laptop	80000	1
2	Laptop B	Laptop	75000	2
3	Laptop C	Laptop	75000	3
5	Phone A	Phone	50000	1
6	Phone B	Phone	45000	2
7	Phone C	Phone	45000	3


```

WITH RECURSIVE Calendar AS (
    SELECT DATE('2025-01-01') AS dt

    UNION ALL

    SELECT DATE(dt, '+1 day')
    FROM Calendar
    WHERE dt < DATE('2025-01-31')
)
SELECT * FROM Calendar;

```

Output

dt
2025-01-01
2025-01-02
2025-01-03
2025-01-04
2025-01-05
2025-01-06
2025-01-07
2025-01-08
2025-01-09

```

CREATE TABLE Products (
    ProductID INT PRIMARY KEY,
    ProductName VARCHAR(50),
    Price DECIMAL(10,2)
);
INSERT INTO Products VALUES
(1, 'Laptop', 800.00),
(2, 'Mouse', 20.00),
(3, 'Keyboard', 40.00);
CREATE TABLE StagingProducts (
    ProductID INT PRIMARY KEY,
    ProductName VARCHAR(50),
    Price DECIMAL(10,2)
);
INSERT INTO StagingProducts VALUES
(2, 'Mouse', 25.00),      -- Existing product (update)
(3, 'Keyboard', 45.00),  -- Existing product (update)
(4, 'Monitor', 200.00);  -- New product (insert)

```

Output

SQL query successfully executed. However, the result set is empty.

```
UPDATE Products
SET Price = (
    SELECT s.Price
    FROM StagingProducts s
    WHERE s.ProductID = Products.ProductID
)
WHERE ProductID IN (
    SELECT ProductID
    FROM StagingProducts
);
```

Output

SQL query successfully executed. However, the result set is empty.

```
INSERT INTO Products (ProductID, ProductName, Price)
SELECT s.ProductID, s.ProductName, s.Price
FROM StagingProducts s
WHERE s.ProductID NOT IN (
    SELECT ProductID
    FROM Products
);
```

Output

SQL query successfully executed. However, the result set is empty.

```
SELECT * FROM Products;
```

Output

ProductID	ProductName	Price
1	Laptop	800
2	Mouse	25
3	Keyboard	45
4	Monitor	200

```
CREATE TABLE Sale (  
    Product TEXT,  
    Month TEXT,  
    Quantity INTEGER  
);  
INSERT INTO Sale VALUES  
( 'Laptop', 'Jan', 10),  
( 'Laptop', 'Feb', 15),  
( 'Laptop', 'Mar', 20),  
( 'Mouse', 'Jan', 30),  
( 'Mouse', 'Feb', 25),  
( 'Mouse', 'Mar', 35);  
SELECT  
    Product,  
    SUM(CASE WHEN Month = 'Jan' THEN Quantity ELSE 0 END) AS Jan,  
    SUM(CASE WHEN Month = 'Feb' THEN Quantity ELSE 0 END) AS Feb,  
    SUM(CASE WHEN Month = 'Mar' THEN Quantity ELSE 0 END) AS Mar  
FROM Sale  
GROUP BY Product;
```

Output

Product	Jan	Feb	Mar
Laptop	10	15	20
Mouse	30	25	35

```

WITH Pivoted AS (
    SELECT
        Product,
        SUM(CASE WHEN Month = 'Jan' THEN Quantity ELSE 0 END) AS Jan,
        SUM(CASE WHEN Month = 'Feb' THEN Quantity ELSE 0 END) AS Feb,
        SUM(CASE WHEN Month = 'Mar' THEN Quantity ELSE 0 END) AS Mar
    FROM Sales
    GROUP BY Product
)
SELECT Product, 'Jan' AS Month, Jan AS Quantity
FROM Pivoted
UNION ALL
SELECT Product, 'Feb', Feb
FROM Pivoted
UNION ALL
SELECT Product, 'Mar', Mar
FROM Pivoted
ORDER BY Product, Month;

```

Output

Product	Month	Quantity
Laptop	Feb	15
Laptop	Jan	10
Laptop	Mar	20
Mouse	Feb	25
Mouse	Jan	30
Mouse	Mar	35

```

CREATE TABLE Departments (
    DepartmentID INTEGER PRIMARY KEY,
    DepartmentName TEXT
);
CREATE TABLE Employees (
    EmployeeID INTEGER PRIMARY KEY,
    FirstName TEXT,
    LastName TEXT,
    DepartmentID INTEGER,
    FOREIGN KEY (DepartmentID) REFERENCES Departments(DepartmentID)
);
INSERT INTO Departments VALUES
(1, 'HR'),
(2, 'IT'),
(3, 'Finance');
INSERT INTO Employees VALUES
(101, 'John', 'Smith', 1),
(102, 'Alice', 'Brown', 2),
(103, 'David', 'Wilson', 3);

```

Output

SQL query successfully executed. However, the result set is empty.

```
CREATE VIEW vw_EmployeeBasicInfo AS
SELECT
    e.EmployeeID,
    e.FirstName,
    e.LastName,
    d.DepartmentName
FROM Employees e
JOIN Departments d
    ON e.DepartmentID = d.DepartmentID;
```

Output

SQL query successfully executed. However, the result set is empty.

```
SELECT * FROM vw_EmployeeBasicInfo;
```

Output

EmployeeID	FirstName	LastName	DepartmentName
101	John	Smith	HR
102	Alice	Brown	IT
103	David	Wilson	Finance

```

CREATE VIEW vw_EmployeeFullName AS
SELECT
    e.EmployeeID,
    e.FirstName,
    e.LastName,
    e.FirstName || ' ' || e.LastName AS FullName,
    d.DepartmentName
FROM Employees e
JOIN Departments d
    ON e.DepartmentID = d.DepartmentID;

```

Output

SQL query successfully executed. However, the result set is empty.

```

SELECT * FROM vw_EmployeeFullName;

```

Output

EmployeeID	FirstName	LastName	FullName	DepartmentName
101	John	Smith	John Smith	HR
102	Alice	Brown	Alice Brown	IT
103	David	Wilson	David Wilson	Finance


```

CREATE TABLE DeptMaster (
    DeptID INTEGER PRIMARY KEY,
    DeptName TEXT
);
CREATE TABLE Staff (
    StaffID INTEGER PRIMARY KEY,
    FName TEXT,
    LName TEXT,
    MonthlySalary REAL,
    DeptID INTEGER
);
INSERT INTO DeptMaster VALUES
(1, 'HR'),
(2, 'IT'),
(3, 'Finance');
INSERT INTO Staff VALUES
(101, 'John', 'Smith', 5000, 1),
(102, 'Alice', 'Brown', 6500, 2),
(103, 'David', 'Wilson', 7000, 3);
CREATE VIEW vw_StaffAnnualPay AS

```

Output

StaffID	FName	LName	MonthlySalary	AnnualSalary
101	John	Smith	5000	60000
102	Alice	Brown	6500	78000
103	David	Wilson	7000	84000

```

(3, 'Finance');
INSERT INTO Staff VALUES
(101, 'John', 'Smith', 5000, 1),
(102, 'Alice', 'Brown', 6500, 2),
(103, 'David', 'Wilson', 7000, 3);
CREATE VIEW vw_StaffAnnualPay AS
SELECT StaffID, FName, LName,
    MonthlySalary,
    MonthlySalary * 12 AS AnnualSalary
FROM Staff;

SELECT * FROM vw_StaffAnnualPay;

```

```

CREATE TABLE Department(DepartmentID INT PRIMARY KEY, DepartmentName VARCHAR(100));
CREATE TABLE Employees (EmployeeID INT PRIMARY KEY, FirstName VARCHAR(50), LastName VARCHAR(50),
DepartmentID INT, Salary DECIMAL(10,2), JoinDate DATE);
INSERT INTO Department VALUES(1,'HR'),(2,'Finance'),(3,'IT'),(4,'Marketing');
INSERT INTO Employees VALUES
(1,'John','Doe',1,5000.00,'2020-01-15'),
(2,'Jane','Smith',2,6000.00,'2019-03-22'),
(3,'Michael','Johnson',3,7000.00,'2018-07-30'),
(4,'Emily','Davis',4,5500.00,'2021-11-05');
SELECT * FROM Employees;

```

Output

EmployeeID	FirstName	LastName	DepartmentID	Salary	JoinDate
1	John	Doe	1	5000	2020-01-15
2	Jane	Smith	2	6000	2019-03-22
3	Michael	Johnson	3	7000	2018-07-30
4	Emily	Davis	4	5500	2021-11-05

```
SELECT * FROM Employees WHERE DepartmentID = 1;
```

Output

EmployeeID	FirstName	LastName	DepartmentID	Salary	JoinDate
1	John	Doe	1	5000	2020-01-15

```
CREATE TABLE Departments ( DepartmentID INTEGER PRIMARY KEY, DepartmentName TEXT);
CREATE TABLE Employees (EmployeeID INTEGER PRIMARY KEY, FirstName TEXT, LastName
TEXT, DepartmentID INTEGER, Salary REAL, JoinDate TEXT, FOREIGN KEY (DepartmentID)
REFERENCES Departments(DepartmentID));
INSERT INTO Departments VALUES (1,'HR'), (2,'IT'), (3,'Finance');
INSERT INTO Employees VALUES
(1,'John','Doe',1,5000.00,'2020-01-15'),
(2,'Jane','Smith',2,6000.00,'2019-03-22'),
(3,'Bob','Johnson',3,5500.00,'2021-07-01');
SELECT EmployeeID, FirstName, LastName, Salary, Salary * 12 AS AnnualSalary
FROM Employees;
SELECT * FROM Employees WHERE DepartmentID = 2;
SELECT EmployeeID, FirstName, Salary, Salary * 0.10 AS Bonus FROM Employees;
SELECT EmployeeID, FirstName, Salary, Salary * 0.15 AS Bonus FROM Employees;
```

Output

EmployeeID	FirstName	LastName	Salary	AnnualSalary
1	John	Doe	5000	60000
2	Jane	Smith	6000	72000
3	Bob	Johnson	5500	66000

EmployeeID	FirstName	LastName	DepartmentID	Salary	JoinDate
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```

(3, 'Bob', 'Johnson', 3, 5500.00, '2021-07-01');
SELECT EmployeeID, FirstName, LastName, Salary, Salary * 12 AS AnnualSalary
FROM Employees;
SELECT * FROM Employees WHERE DepartmentID = 2;
SELECT EmployeeID, FirstName, Salary, Salary * 0.10 AS Bonus FROM Employees;
SELECT EmployeeID, FirstName, Salary, Salary * 0.15 AS Bonus FROM Employees;
SELECT EmployeeID, FirstName, Salary * 12 AS AnnualSalary FROM Employees;
SELECT EmployeeID, FirstName, Salary * 12 AS AnnualSalary FROM Employees WHERE
EmployeeID = 1;
SELECT * FROM Employees WHERE DepartmentID = 3;
SELECT EmployeeID, FirstName, Salary, Salary * 12 AS AnnualSalary, Salary * 0.15 AS
Bonus, (Salary * 12) + (Salary * 0.15) AS TotalCompensation FROM Employees;
SELECT EmployeeID, FirstName, Salary, Salary * 12 AS AnnualSalary, Salary * 0.15
AS Bonus, (Salary * 12) + (Salary * 0.15) AS TotalCompensation FROM Employees;

```

Output

2	Jane	Smith	2	6000	2019-03-22
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EmployeeID	FirstName	Salary	Bonus
1	John	5000	500
2	Jane	6000	600
3	Bob	5500	550

EmployeeID	FirstName	Salary	Bonus
1	John	5000	750
2	Jane	6000	900
3	Bob	5500	825

EmployeeID	FirstName	AnnualSalary
1	John	60000

2	Jane	72000
3	Bob	66000

EmployeeID	FirstName	AnnualSalary
1	John	60000

EmployeeID	FirstName	LastName	DepartmentID	Salary	JoinDate
------------	-----------	----------	--------------	--------	----------

3	Bob	Johnson	3	5500	2021-07-01
EmployeeID	FirstName	Salary	AnnualSalary	Bonus	TotalCompense
1	John	5000	60000	750	60750
2	Jane	6000	72000	900	72900
3	Bob	5500	66000	825	66825

EmployeeID	FirstName	Salary	AnnualSalary	Bonus	TotalCompense
1	John	5000	60000	750	60750
2	Jane	6000	72000	900	72900
3	Bob	5500	66000	825	66825